

An Anatomy of the Volatility of Cryptocurrency: Evidence from Ethereum and DAO Hack Event*

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Abstract

The cryptocurrency market recently gained a lot of attention from investors. But, its volatility has been acting as a disincentive to investment. Volatility plays an important role in shaping market riskiness and investment behavior. We study the volatility of the Ethereum (ETH) cryptocurrency from the following perspectives. The first goal of this study is to identify risk-seeking behavior in the ETH cryptocurrency market. We examine this propensity by measuring the effect of the volatility of Ethereum on the total ETH assets. This investigation also takes the form of a case-study of an unexpected ETH fund-stolen event, DAO Hack, and the hard fork treatment. We also forecast a downward volatility trend in the near future based on Autoregressive models. This is the first study to analyze DAO Hack with empirical methods and marks the starting point for more rigorous models to predict the volatility of Ethereum.

Keywords: Cryptocurrency Market, Regression Discontinuity, Autoregression

JEL Codes: E42, G10, G17

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1 Introduction

Recent advances in decentralized finance (Defi) on the Ethereum blockchain have provided economists with a envision of future financial systems. Ethereum is an open-source blockchain-based platform that creates and shares business, financial services, and entertainment applications with its own cryptocurrency, Ether, or ETH (Buterin, 2013). During the last three years, we have seen a robust increase in the ETH price and total assets.

Although Ethereum has the potential for extraordinary returns, it is also a risky investment. Barchart Technical Analysis (2021) shows that the price volatility of Ethereum is 131.45% by 2021/6/26. Volatility is one of the most important indexes in both stock and cryptocurrency exchange markets, which measures the variance of the rate of return. Bhowmik and Wang (2020) particularly argues that volatility is directly related to market uncertainty and it affects investment behavior, hence the demand dynamics of the market.

Recent works have begin to study the volatility of cryptocurrency (Katsiampa, P., 2017; Baur and Dimpfl, 2018; Kyriazis et al., 2019). However, no previous research has investigated about the volatility of Ethereum with econometric analysis. In our research, we analyze the Ethereum volatility through the following three aspects: 1). Examining the role of volatility in affecting the ETH total investment (demand). 2). Case-study: the local effect of the DAO hack event (unexpected ETH fund-stolen event) on the ETH volatility and the role of subsequent hard fork treatment on recovering the ETH volatility. 3). Forecasting the ETH volatility. To achieve the objective, we propose four econometric models, including Two-Stage Least Squares (2SLS), Regression Discontinuity, Difference-in-Differences (DiD), and p th-order Autoregressive model. Our strategy for the first objective is using a supply-side instrument to cope with the endogeneity issue of ETH volatility. The OLS and 2SLS results uncover a positive effect of volatility on the total ETH asset demand. In the DAO Hack case-study, we apply Regression Discontinuity design and find that the local effect of the DAO Hack event does not cause a leap in ETH volatility at the cutoff date. Still, Difference-in-Difference method unveils that the hard fork treatment by ETH officials generates a pacifying

impact on ETH volatility. Finally, we forecast the volatility of ETH based on its historical data and argue that the volatility will go down in the near future.

In section 2, we introduce the two data sources for our analysis. In section 3, we discuss the DAO Hack background and detailed motivations for the empirical methodologies. We also review the properties and previous examples of the Econometric models to clarify their fitness for our investigation objectives. In section 4, we interpret the empirical results. Section 5 concludes.

2 Data

Our empirical analysis is based on two sets of data sources. One of the datasets is the Coin Metrics Data. Table 1 ¹presents detailed description for the variables that we are using for volatility investigation.

ID	Name	Description
VtyDayRet30d	Volatility, daily returns, 30d	The 30D volatility, measured as the standard deviation of the natural log of daily returns over the past 30 days.
TxTfrValAdjUSD	Transactions, transfers, value, adjusted, USD	The USD value of the sum of native units transferred between distinct addresses that interval removing noise and certain artifacts.
PriceUSD	Price, USD	The fixed closing price of the asset as of 00:00 UTC the following day (i.e., midnight UTC of the current day)

¹Open source data from the Coin Metrics is applied and imported via Pandas through Python. Reference URL: <https://github.com/coinmetrics-io/data/blob/master/csv/metrics.csv>

		denominated in USD. This price is generated by Coin Metrics' fixing/reference rate service. Real-time PriceUSD is the fixed closing price of the asset as of the timestamp set by the block's miner.
AdrBalCnt	Addresses, any balance, count	The sum count of unique addresses holding any amount of native units as of the end of that interval. Only native units are considered (e.g., a 0 ETH balance address with ERC-20 tokens would not be considered).
NVTAdj90	NVT, adjusted, 90d MA	The ratio of the free float network value (or market capitalization, free float) to the 90-day moving average of the adjusted transfer value. Also referred to as FFNVT.
SplyCur	Supply, current	The sum of all native units ever created and visible on the ledger (i.e., issued) at the end of that interval. For account-based protocols, only accounts with positive balances are counted.
FeeMedUSD	Fees, transaction, median, USD	The USD value of the median fee per transaction that interval.
RevUSD	Miner revenue, USD	The sum USD value of all miner revenue (fees plus newly issued native units) that interval.
GasLmtTxMean	Gas, limit, transaction, mean	The mean gas limit per transaction that interval.

Table 1: Coin Metrics Data Information (Jiarui, 2021)

The other data source is the DeFi Pulse Data which provides us with the Total Value Locked Data for Ethereum over a given period of time. Total Value Locked (TVL) represents the number of assets that are currently being staked in a specific protocol, which measures the total assets value of ETH platform (Stepanova, 2021). Table 2 ² includes all the TVL data that are used in our analysis.

²To import the historical data of TVL, we create a free account at data.defipulse.com to generate an API key. Reference URL: <https://docs.defipulse.com/api-docs-by-provider/defi-pulse-data/total-value-locked/total-value-locked>

ID	Description
TVLUSD	The USD value of the number of assets that are currently being staked in a specific protocol
TVLETH	The Ether value of the number of assets that are currently being staked in a specific protocol
Timestamp	A timestamp is temporal information regarding an event that is recorded by the computer and then stored as metadata. Any event or activity could have a timestamp recorded. Here, we convert the timestamp into Date Time.

Table 2: TVL Data Information (Jiarui, 2021)

3 Motivation and Literature Review

In this section, we further discuss the ETH and DAO Hack background and elucidate the motivation for us to implement corresponding econometric models. As we do not trace much economics literature about ETH and DAO Hack so far, we will review some previous paragons on the econometric models.

3.1 Background and Motivation

The effect of the volatility of return on investment is hard-tracking in financial markets because of its endogeneity with changes in other markets, interest rates, monetary policy, etc. ETH volatility, which is interpreted as the variation of the differentials between the open price and close price, exhibits a price endogeneity issue as well. We need a sensible instrument variable to isolate the effect of the volatility of return on ETH volume or demand. MacKay and Miller (2015) particularly argue that a supply-side instrument could resolve the price endogeneity in demand estimation. Henceforth, we apply ETH-issued units (or supply) as the instrument variable and estimate the effect of the volatility of return on the total ETH

assets. As the total ETH assets variable identifies the investment volume, such impact reveals the risk propensity of Ethereum investors

DAO is a decentralized autonomous organization implemented on Ethereum blockchain technology. Investors add funds into DAO by purchasing ETH tokens and obtaining ownership in order to get involved in operation decision and investment strategy. On 16 June 2016, an attacker managed to withdraw approximately 3.6 million Ether from the DAO fund by abusing a coding loophole. This attack event, namely DAO Hack, incurred direct losses to the ETH investors and sparked a huge sell-off. For a stock market crash, Mazur et al. (2021) argue that stocks exhibit extreme asymmetric volatility that correlates negatively with stock returns. In a similar direction but a different institution, we want to examine how ETH cryptocurrency volatility responds to a sudden asset loss in DAO fund. The result will be meaningful to understanding the sensitivity of financial markets' volatility to shocks and to producing further research on risk-seeking individuals' investment behavior toward sudden drastic loss, in line with Lionel (2014).

After the DAO Hack, Ethereum founders effectively proposed a hard fork to roll back the ETH network's history, which technically returned the lost fund to the investors. ETH officials kept a pre-forked version — now known as Ethereum Classic (ETC) for the asset holders who refused the hard fork. The coexistence of hard-forked version of ETH and pre-forked version of ETH offers a perfect identification for us to investigate the treatment effect of the hard fork. Therefore, we conduct a Difference in Difference design to understand the remedial effect of a "policy experiment".

For forecasting volatility in financial markets, Poon (2003) examines 93 relevant pieces of literature and summarizes two main approaches: historical volatility models and volatility implied from options. TG Andersen (2005), RF Engle (2007), and Tim (2017) continued adding to this strand of theoretical literature and advancing the accuracy of volatility. Nonetheless, we find a few articles talking about ETH volatility prediction. We intend to exploit Autoregressive Model and check the performance of using the historical data to predict

future ETH volatility.

3.2 Applicability

3.2.1 Two-Stage Least Squares(2SLS)

2SLS is one of the most commonly used techniques and the extension of the OLS function, which is typically introduced as solutions to the problem of “simultaneous equations bias”, where dependent variable’s error terms are correlated with the independent variables. (Joshua, 1995) According to the Gauss-Markov theorem, the OLS estimate for regression coefficients gives us the best linear unbiased estimate (BLUE) if five assumptions are met as shown in Figure 2. When x is correlated with the error term ε , assumption 2 of the Gauss-Markov Theorem is violated. An instrument variable z is adopted and a new assumption is added: z is correlated with x but not correlated with ε . This assumption isolates the movements in x that are uncorrelated with ε . We can get the BLUE estimator by the Gauss-Markov theorem again as shown in Figure 3.

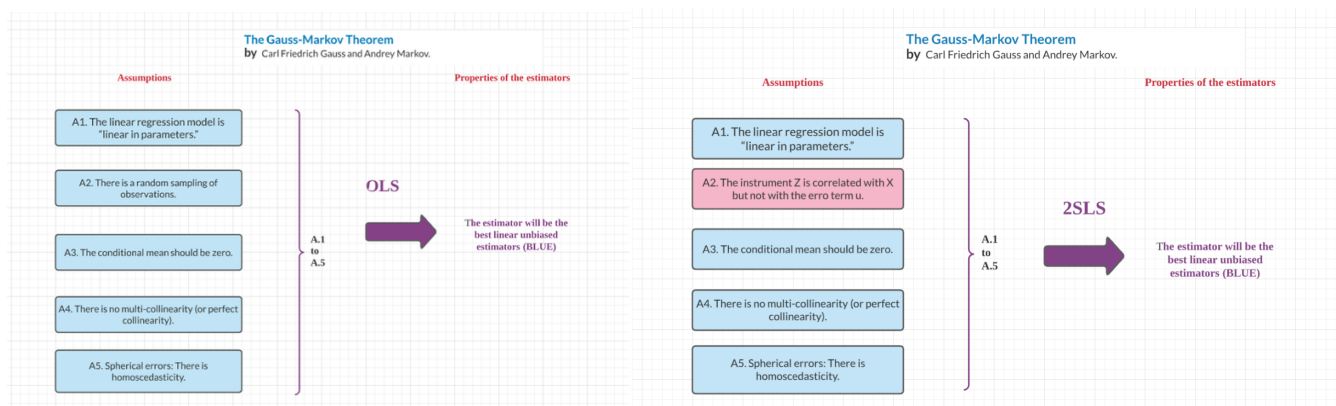


Figure 1: OLS (Jiarui, 2021)

Figure 2: 2SLS (Jiarui, 2021)

3.2.2 Regression Discontinuity

The Regression Discontinuity method works particularly well in a situation when the treatment variable D_i is solely determined by whether a continuous variable x_i is above or below

a cutoff point.

$$D_i = \begin{cases} 1 & \text{if } x \geq 1000 \\ 0 & \text{otherwise} \end{cases}$$

If we assume pretreatment y_i has a linear relationship with x_i : $y_i = \alpha + \beta x_i + \varepsilon_i$. D_i is a function of x_i , noted as $D(x_i)$. We can estimate the potential causal effect of $D(x_i)$ on y_i . We can regard x near 1000 showing no systematic difference. In other words, random sampling incurs that $x = 1000 + \epsilon$ are in the treatment group and $x = 1000 - \epsilon$ are in the control group. we can estimate the Local Average Treatment Effect (LATE):

$$LATE \equiv E(y_{1i}|x = 1000) - E(y_{0i}|x = 1000)$$

If $D_i = 1$, the linear relationship exists a leap at the cutting point ($x = 1000$), which is considered as the causal effect of D_i to y_i . To examine the LATE or the cutoff leap, we conduct OLS with the following equation and identify $\hat{\delta}$:

$$y_i = \alpha + \beta(x_i - 1000) + \delta D_i + \gamma(x_i - 1000)D_i + \varepsilon_i$$

3.2.3 Difference-in-Difference

Difference-in-Difference (DiD) method is useful for capturing the difference between the explained variable before and after a natural experiment. A typical 2-period panel model could be:

$$y_{it} = \beta_0 + \beta_1 G_i \cdot D_t + \beta_2 G_i + \gamma D_t + \varepsilon_{it}$$

where $G_i = 1$ if i belongs to the treatment group and $G_i = 0$ if i belongs to the control group. $D_t = 1$ if $t = 2$ and $D_t = 0$ if $t = 1$. G_i identifies the initial difference between control and treatment groups and D_t measures the time trend that is universal among both groups. Thus, the interaction variable $G_i \cdot D_t$ captures the real treatment effect of the natural experiment.

3.2.4 AR(p)

The Autoregressive model has the following property: When there exists time-series data when y_t is linearly correlated with y_{t-1} , the Homoskedasticity assumption of OLS is violated. However, if y_t is correlated with y_{t-1} but not y_{t-p} for $p > 1$, then we can imply that by running the following regression, i.e., AR(1), we can get the BLUE estimator according to the Gauss-Markov theorem. The inference generally applies to AR(p) models with up to p lag terms.

$$y_t = \alpha + \beta_1 y_{t-1} + \cdots + \beta_p y_{t-p} + \varepsilon_t$$

3.3 Previous Applications

A classic example of 2SLS is Joshua's work on education return (1995). OLS and 2SLS are implemented to estimate the effect of compulsory school attendance on earnings. The dependent variable is earnings or return and the independent variable is education. However, it is commonly known that disparity in earnings is highly likely due to the difference in individuals' capacity. Thus, Joshua adopts an instrument variable, three-quarters of birth, into the study. From figure 3, we can see that 2SLS is implemented to overcome the endogeneity and obtain a more unbiased result than OLS.

	<i>Born 1920–1929, 1970 census</i>				<i>Born 1930–1939, 1980 census</i>			
	<i>OLS</i>	<i>TSLS</i>	<i>OLS</i>	<i>TSLS</i>	<i>OLS</i>	<i>TSLS</i>	<i>OLS</i>	<i>TSLS</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Education	.080 (.0004)	.063 (.016)	.080 (.0004)	.077 (.015)	.071 (.0003)	.103 (.020)	.071 (.0003)	.089 (.016)
YOB dummies	no	no	yes	yes	no	no	yes	yes
$\chi^2(\text{dof})$		2.4 (2)		36.0 (29)		2.9 (2)		25.4 (29)
Sample size		247,199				329,509		

Figure 3: Return to Schooling (Joshua, 1995)

Leah (2017) exploits regression discontinuity to examine how a prescription drug program's free offers to those with an annual income of less than \$20,000 influence the health conditions of these people. Figure 4(c) illustrates a discontinuous health condition at the

income cut-off. While those who earned marginally lower than \$20,000 receive the free drug and perceive 3 percentage points higher in health conditions than the people who earned marginally higher than \$20,000.

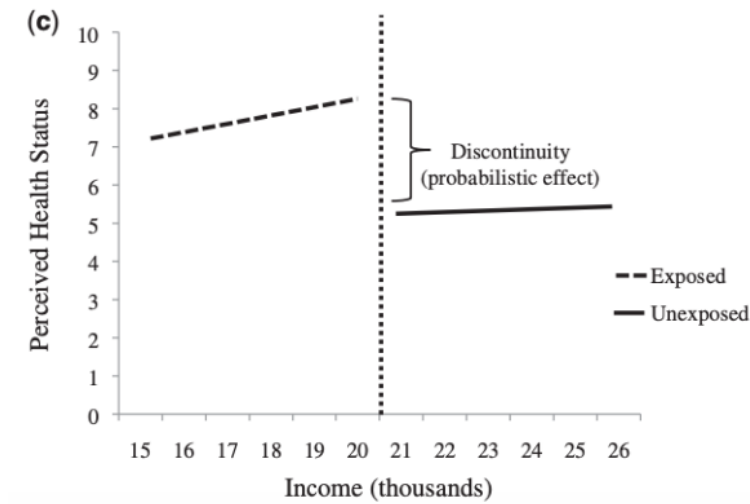


Figure 4: Free Drug Provision at Cutoff to Health Condition (Leah, 2017)

Gruber (1994) conducts a DiD study focusing on whether women pay for the newly mandated childbirth coverage indirectly via a pay cut. The researcher compared young, married women's wages before and after the mandate. We can see from figure 5 that young and married women's wages decreased by 6.2% due to the childbirth coverage mandate.

Location/year	Before law change	After law change	Time difference for location
A. Treatment Individuals: Married Women, 20–40 Years Old:			
Experimental states	1.547 (0.012) [1,400]	1.513 (0.012) [1,496]	–0.034 (0.017)
Nonexperimental states	1.369 (0.010) [1,480]	1.397 (0.010) [1,640]	0.028 (0.014)
Location difference at a point in time:	0.178 (0.016)	0.116 (0.015)	
Difference-in-difference:			–0.062 (0.022)

Figure 5: Childbirth Coverage Mandate to Women Wage Difference (Gruber, 1994)

4 Regression Result

4.1 OLS and IV2SLS: ETH Volatility on ETH Asset Demand

As discussed in section 3, ETH volatility (variation of ETH trading price series over a period of time) is highly likely to be correlated with the variables in the error term, ε . So, we adopt the supply (SplyCur) as the instrumental variable to isolate the movements in volatility in order to permit consistent estimation of the regression coefficient in accordance with MacKay (2015). Besides, we set the Total ETH Asset (TVL) for Ethereum Blockchain to be the dependent variable, which could be regarded as the demand for ETH assets from traders. Since ETH investment is widely considered gambling with respect to daily volatility, we conduct our demand analysis with medium-term 30-Day Volatility (VtyDayRet30d) to be the independent variable. The sum of address numbers on ETH (AdrBalCnt) is the other control variable. The regression equation that we use is:

$$Y_i = \beta_0 + \beta_1 \cdot X_{it} + W_{it} + \varepsilon_{it}$$

Then, IV is validated as Figure 6 shows that the IV (SplyCur) is negatively correlated with the endogenous variable 30-Day Volatility.

	VtyDayRet30d	SplyCur
VtyDayRet30d	1.000000	-0.101702
SplyCur	-0.101702	1.000000

Figure 6: Correlation: 30-Day Volatility and Current Supply (Jiarui, 2021)

We then regress the TVLUSD with both OLS and 2SLS. Figure 8 represents the first-stage negative relationship between supply and volatility. Figure 7 shows that, by 2SLS estimation, the Total ETH Asset increases by $1.91 \cdot 10^{12}$ for a one-unit increase in 30-Day Volatility. Meanwhile, the OLS regression indicates that the TVL increases by $7.16 \cdot 10^{10}$ for a one-unit increase in 30-Day Volatility. The coefficients display huge differences between the OLS

Model Comparison		
	OLS	2SLS
Dep. Variable	tv1USD	tv1USD
Estimator	OLS	IV-2SLS
No. Observations	1417	1417
Cov. Est.	robust	robust
R-squared	0.7346	-3.5537
Adj. R-squared	0.7341	-3.5601
F-statistic	1383.4	219.83
P-value (F-stat)	0.0000	0.0000
const	6.909e+11 (27.410)	-1.157e+11 (-9.1159)
VtyDayRet30d	7.16e+10 (5.5755)	1.916e+12 (8.1341)
SplyCur	-7491.6 (-27.935)	
AdrBalCnt	3622.7 (30.134)	859.39 (14.055)
Instruments		SplyCur

First Stage Estimation Results	
	VtyDayRet30d
R-squared	0.0577
Partial R-squared	0.0530
Shea's R-squared	0.0530
Partial F-statistic	68.631
P-value (Partial F-stat)	1.11e-16
Partial F-stat Distn	chi2(1)
const	0.4374 (9.4354)
AdrBalCnt	1.498e-09 (7.8103)
SplyCur	-4.062e-09 (-8.2844)

Figure 7: OLS and 2SLS Comparison (Jiarui, 2021) Figure 8: First Stage Result (Jiarui, 2021)

regression and 2SLS Regression results. Negative R^2 reveals that the 2SLS model prediction performs worse than the mean of the target values, thus demonstrating less reliability. On the other hand, we reject the exogenous null hypothesis in Durbin test and conclude that OLS estimates will be relatively efficient. To sum up, we interpret that higher ETH volatility of return (price differentials) over 30-day periods positively impact the total assets on the ETH blockchain. In other words, the riskier the return, the higher the investment in ETH. This subtly indicates that individuals investing in the ETH cryptocurrency market are more likely to be risk-seeking. This implication is consistent with Matthias Pelster's article (2019) which argues that cryptocurrency traders are motivated by risk-seeking behavior.

4.2 Regression Discontinuity: DAO Hack

The DAO Hack event brought about a direct loss in ETH token investors' assets, thus undermining the reliability of crypto market institutions and giving rise to a plummet in ETH price from \$21 to \$12. Even though the ETH price dropped significantly, the sensitivity of volatility to such a market crash is ambiguous. In particular, we plan to see if there exists

a discontinuous leap at the DAO Hack cutoff date (16 Jun 2016).

$$AfterDAOHack = \begin{cases} 1 & \text{if } x \geq date(2016, 6, 16) \\ 0 & \text{elsewise} \end{cases}$$

We import the datetime variable and create the identifier variable (*AfterDAOHack*) which is solely determined by whether the date is before or after the DAO Hack event. Then, we normalize the independent (days) variable to be the number of days before or after the event ($date - 6/16/2016$) and derive the following regression equation:

$$Y_i = \alpha + \beta days + \delta AfterDAOHack + \gamma(days \cdot AfterDAOHack) + X_i + \varepsilon_i$$

OLS Regression Results						
=====						
Dep. Variable:	VtyDayRet30d	R-squared:	0.250			
Model:	OLS	Adj. R-squared:	0.248			
Method:	Least Squares	F-statistic:	174.9			
Date:	Wed, 16 Jun 2021	Prob (F-statistic):	2.01e-129			
Time:	07:16:36	Log-Likelihood:	5097.1			
No. Observations:	2108	AIC:	-1.018e+04			
Df Residuals:	2103	BIC:	-1.016e+04			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.0683	0.002	28.122	0.000	0.064	0.073
After_DaoHack	-0.0091	0.003	-3.436	0.001	-0.014	-0.004
days	-9.438e-05	1.32e-05	-7.124	0.000	-0.000	-6.84e-05
After_DaoHack:days	8.493e-05	1.33e-05	6.390	0.000	5.89e-05	0.000
TxFrValAdjUSD	1.043e-12	1.29e-13	8.070	0.000	7.89e-13	1.3e-12
=====						
Omnibus:	313.403	Durbin-Watson:	0.049			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	610.476			
Skew:	0.909	Prob(JB):	2.73e-133			
Kurtosis:	4.909	Cond. No.	3.19e+10			
=====						

Figure 9: Regression Discontinuity Result (Jiarui, 2021)

We implement OLS regression with the above equation and obtain the estimated $\hat{\delta}$ coefficient. The $\hat{\delta}$ shown in figure 9 identifies a statistically significant estimation of the LATE effect on the DAO Hack cutoff date. Nonetheless, the numerical amount of the coefficient is negligible. In other words, there does not exist a discontinuity at the DAO Hack cutoff moment. We conclude that the ETH volatility does not show great sensitivity to a sudden

price drop.

4.3 Difference-in-Difference: Treatment Effect of ETH Hard Fork

After the DAO Hack, Ethereum officials proposed a hard fork solution for the bug and managed to roll back and return the lost fund. In this natural experiment, we find the perfect treatment group, hard-forked ETH, and control group ETH classic (ETC), the original unforked (or attacked) ETH that was kept in the ETH community. In the book “Out of the Ether”, Matthew strongly believes that this treatment solution recovered the ETH’s reliability and investment, hence saving the Ethereum blockchain. We want to utilize the DiD design to capture to what extent the remedial treatment affects the return volatility of ETH in comparison to ETC.

$$etc = \begin{cases} 1 & \text{if } type = ETC \\ 0 & \text{if } type = ETH \end{cases}$$

We construct the dummy variable (*etc*) in the above fashion and derive the following regression equation:

$$Y_i = \alpha + \beta days + \gamma etc + \delta(days \cdot etc) + X_i + \varepsilon_i$$

where β represents the time trend common to ETH and ETC; γ refers to the treatment group-specific effect; δ reveals the true effect of treatment.

From figure 10, a high R^2 shows the suitability of DiD model here to explain the data. We argue that the hard fork indeed causes a difference between ETH and ETC volatility of return in the .01 level. The estimated $\hat{\delta}$ displays a positive direction, saying that the ETC platform shows significantly higher volatility of return than the hard-forked ETH platform. This result states that the hard fork pacifies the instability of the cryptocurrency market to a certain degree and is consistent with Matthew’s idea that this treatment does save the ETH from the DAO Hack market crash.

OLS Regression Results						
=====						
Dep. Variable:	VtyDayRet30d	R-squared:	0.091			
Model:	OLS	Adj. R-squared:	0.089			
Method:	Least Squares	F-statistic:	50.43			
Date:	Wed, 16 Jun 2021	Prob (F-statistic):	8.64e-69			
Time:	07:16:36	Log-Likelihood:	8106.2			
No. Observations:	3542	AIC:	-1.620e+04			
Df Residuals:	3534	BIC:	-1.615e+04			
Df Model:	7					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

Intercept	0.0591	0.001	43.713	0.000	0.056	0.062
etc	0.0087	0.002	5.050	0.000	0.005	0.012
days	-8.098e-06	1.29e-06	-6.297	0.000	-1.06e-05	-5.58e-06
etc:days	5.406e-06	1.72e-06	3.137	0.002	2.03e-06	8.79e-06
TxTfrValAdjUSD	2.715e-13	2.33e-13	1.165	0.244	-1.86e-13	7.28e-13
FeeMedUSD	-0.0043	0.001	-7.778	0.000	-0.005	-0.003
RevUSD	9.726e-10	1.16e-10	8.368	0.000	7.45e-10	1.2e-09
GasLmtTxMean	-2.668e-08	3.09e-09	-8.645	0.000	-3.27e-08	-2.06e-08
=====						
Omnibus:	1019.755	Durbin-Watson:	0.073			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3465.452			
Skew:	1.428	Prob(JB):	0.00			
Kurtosis:	6.915	Cond. No.	1.63e+10			
=====						

Figure 10: DiD regression Result (Jiarui, 2021)

4.4 Autoregressive Model: Volatility Forecast

Through building up an autoregressive model for ETH volatility, we find that the 30-Day Volatility of Ethereum is likely to be predictable with the historical data of itself.

4.4.1 Construction of the Stationary Variable

In this section, we start by establishing and validating the stationary variables: the 30-Day Volatility, lag variable, difference variable, and percent change variable.

```

ADF Statistic: -5.791550
p-value: 0.000000
Critical Values:
 1%: -3.434
 5%: -2.863
10%: -2.568

```

Figure 11: The Dickey-Fuller test for stationary: 30-Day Volatility(Jiarui, 2021)

Figure 11 represents the Dickey-Fuller test for stationary on 30-Day Volatility. We can reject the null hypothesis that 30-Day Volatility is non-stationary with a p-value of 0.000.

Therefore, we will use the 30-Day Volatility itself as a regressor in the AR(p) model.

4.4.2 Visualization of Autocorrelation

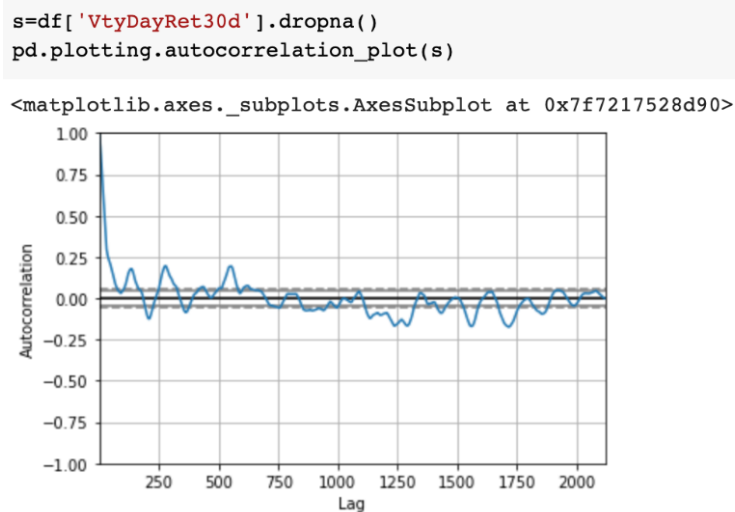


Figure 12: Autocorrelation of the 30-Day Volatility lag variables (Jiarui, 2021)

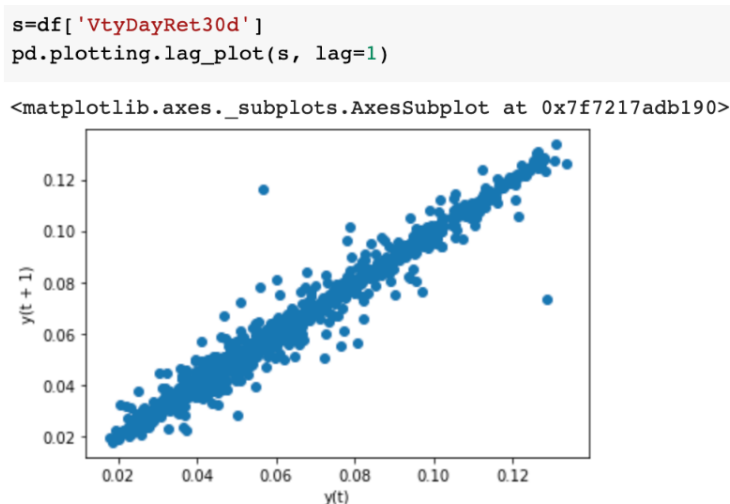


Figure 13: Autocorrelation plot for lag 1: 30-Day Volatility (Jiarui, 2021)

We further illustrate the auto-correlation of 30-Day Volatility. Figure 12 shows the autocorrelation of the 30-Day Volatility lag variables. We can see that the autocorrelation of lag variables is statistically significant with approximately the first 100 lags and then fluctuates near zero correlation. So, we narrow down the lag choice to the first 100 periods. Figure

13 visualizes the autocorrelation plot for the 30-Day Volatility with lag 1, and supplements that there is a very significant linear correlation with lag 1. Thus, the first-order autoregression model could be helpful. To validate, we manually conduct an OLS regression and get a consistent result in figure 14. The lag 1 volatility is a good estimator for the current volatility.

OLS Regression Results						
Dep. Variable:	VtyDayRet30d	R-squared:	0.975			
Model:	OLS	Adj. R-squared:	0.975			
Method:	Least Squares	F-statistic:	6.884e+04			
Date:	Tue, 29 Jun 2021	Prob (F-statistic):	0.00			
Time:	07:36:30	Log-Likelihood:	7578.3			
No. Observations:	1786	AIC:	-1.515e+04			
Df Residuals:	1784	BIC:	-1.514e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.0007	0.000	3.269	0.001	0.000	0.001
VtyDayRet30d_Minus_1	0.9863	0.004	262.381	0.000	0.979	0.994
Omnibus:	783.449	Durbin-Watson:	1.675			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	571288.969			
Skew:	0.544	Prob(JB):	0.00			
Kurtosis:	90.611	Cond. No.	45.8			

Figure 14: AR(1) (Jiarui, 2021)

4.4.3 The Choice of the Number of Lag: BIC and AIC Criteria

AutoReg Model Results					
Dep. Variable:	VtyDayRet30d	No. Observations:	1784		
Model:	AutoReg(2)	Log Likelihood	-1.514e+04		
Method:	Conditional MLE	S.D. of innovations	0.000		
Date:	Sun, 27 Jun 2021	AIC	-1.514e+04		
Time:	02:39:08	BIC	-1.514e+04		
Sample:	2	HQIC	-1.514e+04		
	coef	std err	z	P> z	[0.025
const	0.0008	0.000	3.785	0.000	0.000
VtyDayRet30d.L1	1.1487	0.023	49.158	0.000	1.101
VtyDayRet30d.L2	-0.1642	0.023	-7.034	0.000	-0.210
Roots					
	Real	Imaginary	Modulus	Phase	
AR.1	1.0190	+0.0000j	1.0190	0.0000	
AR.2	5.9753	+0.0000j	5.9753	0.0000	

Figure 15: BIC(i) (Jiarui, 2021)

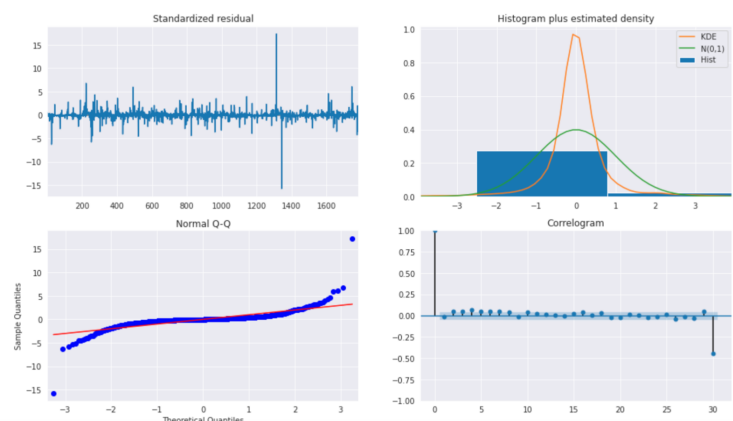


Figure 16: BIC(ii) (Jiarui, 2021)

We also conduct an autoregression with both BIC and AIC criteria to identify the choice of the number of lags. Figure 15 shows the AR(p) regression result using BIC criteria for

the residual diagnosis. The result shows that the lag 1 and 2 terms are significant for BIC criteria, where we can estimate with the AR(2) equation:

$$Y_t = 0.001 + 1.15Y_{t-1} - 0.16Y_{t-2} + \varepsilon_t$$

AutoReg Model Results					
Dep. Variable:	VtyDayRet30d	No. Observations:	7		
Model:	AutoReg(11)	Log Likelihood			
Method:	Conditional MLE	S.D. of innovations			
Date:	Sun, 27 Jun 2021	AIC			
Time:	04:07:53	BIC			
Sample:	11	HQIC			
	1784				
	coef	std err	z	P> z	[0.02
const	0.0012	0.000	5.358	0.000	0.00
VtyDayRet30d.L1	1.1214	0.024	47.274	0.000	1.07
VtyDayRet30d.L2	-0.0999	0.036	-2.803	0.005	-0.17
VtyDayRet30d.L3	-0.0037	0.036	-0.104	0.917	-0.07
VtyDayRet30d.L4	0.0166	0.036	0.463	0.643	-0.05
VtyDayRet30d.L5	-0.0176	0.036	-0.492	0.623	-0.08
VtyDayRet30d.L6	0.0073	0.036	0.204	0.839	-0.06
VtyDayRet30d.L7	0.0013	0.036	0.036	0.971	-0.06
VtyDayRet30d.L8	-0.0170	0.036	-0.472	0.637	-0.08
VtyDayRet30d.L9	-0.0519	0.036	-1.438	0.151	-0.12
VtyDayRet30d.L10	0.0690	0.036	1.914	0.056	-0.00
VtyDayRet30d.L11	-0.0485	0.024	-2.023	0.043	-0.09

Figure 17: AIC(i) (Jiarui, 2021)

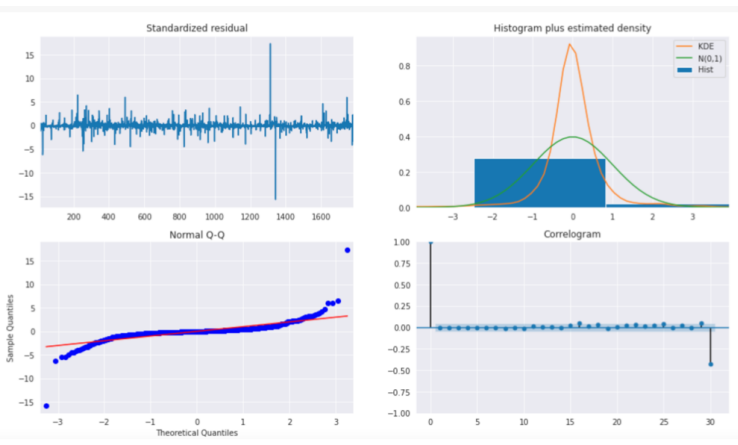


Figure 18: AIC(ii) (Jiarui, 2021)

From the AIC criteria in figure 17, we find a similar result that lag 1, 2, and 11 terms are statistically significant while the other lag coefficients seem to be the white noise. Thus, we still conclude with the AR(2) equation:

$$Y_t = 0.001 + 1.12Y_{t-1} - 0.01Y_{t-2} - 0.05Y_{t-11} + \varepsilon_t$$

From figure 16 and figure 18, the standardized residual graphs are both evenly distributed on the whole and Normal Q-Q graphs also show both sets of quantiles truly come from normal distributions. However, histogram graphs vaguely indicate a normal distribution, which is potentially problematic regarding accuracy.

4.4.4 Prediction

In this subsection, we present the recent trend till today and predict the future trend. Figure 19 presents the prediction plots for the AR(2) that are proven to be useful for prediction.

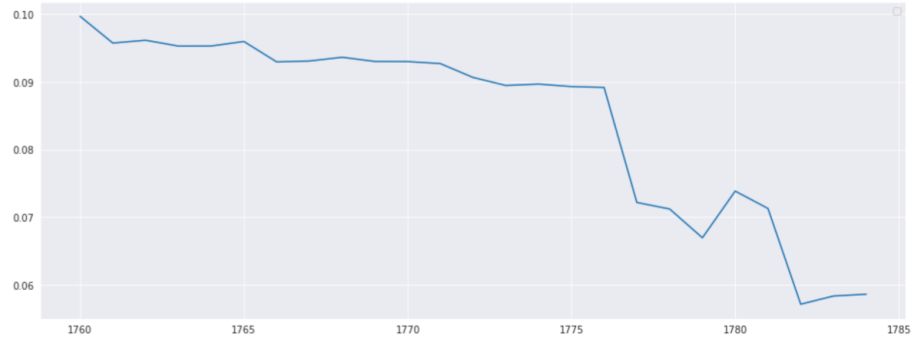


Figure 19: Trend till today(index: 1785) (Jiarui, 2021)

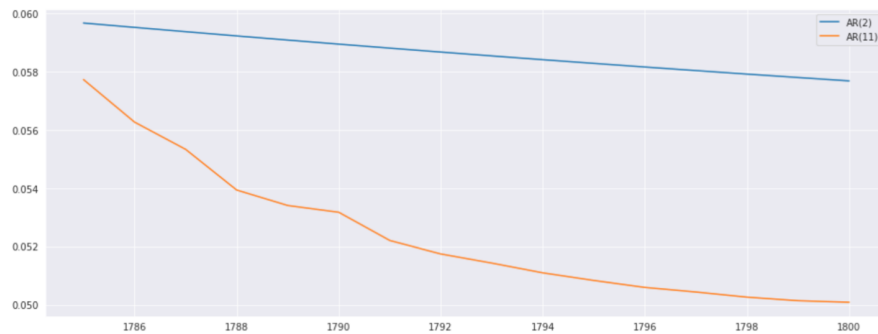


Figure 20: Prediction with AR(2) and AR(11) (Jiarui, 2021)

We also include the AR(11) regression for comparison. We can see that the AR(2) model predicts a smooth and gentle decrease in future volatility while the AR(11) model anticipates a huge drop in the future volatility of Ethereum. They are likely to show more diverge further in the long run. Figure 20 is consistent with the historic trend in figure 19 which shows a general decreasing trend till the last day available in our dataset. Thus, we illustrate that AR(2) and AR(11) display decent performance of prediction, and AR(2) should be more accurately describing the general trend.

5 Conclusion

This paper studies the volatility of Ethereum through three main perspectives: analyzing the role of ETH volatility in the demand for ETH investment. In a perfect natural experiment settings of DAO Hack and hard fork solution, we present negligible cutoff effect but validate

the remedial effect of market-saving treatment. Lastly, we forecast that the ETH volatility will be gently decreasing with historic data.

Our result tells us that higher ETH volatility positively impacts the total assets invested in ETH, thus revealing potential risk-seeking behavior among ETH investors. Although we fail to find volatility discontinuity at the DAO Hack event, we indeed observe the ETH market becomes less volatile due to the hard fork treatment effect. Our next step is studying how risky behavior responds to a large losses due to DAO hack event and the recovery by hard fork treatment.

We anticipate that this paper opens a strand of literature utilizing DAO Hack and hard fork treatment settings for further research. We also hope the autoregressive model we use marks a starting point for more rigorous time-series models in order to better predict the volatility of the financial market, especially that of the cryptocurrency market.

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